CHAARGOLD

Crafting Carbon Perfection



1 TON

OUTPUT CAPACITY

PRODUCT BROCHURE · KILN SERIES

Pyrolysis & *Retort* Systems

Precision-engineered carbonization technology that converts biomass into premium charcoal, biochar and wood vinegar — built in-house, built to last.

ENGINEERING & DESIGN

Engineered for Excellence

Manufactured in-house by our team of engineers and technical specialists, every ChaarGold kiln is designed to optimize the pyrolysis process through precise temperature control, efficient heat utilization and enhanced carbon recovery.

> *Biomass in, carbon-rich value out — every cycle, with less fuel, fewer emissions and lower operating cost.*

Pyrolysis heats biomass in an oxygen-limited chamber so that it carbonizes instead of burning. Done well, the result is a dense, high-grade carbon product, along with condensable vapours that can be captured and sold. Our systems are built to deliver exactly that — consistently, batch after batch.



GANTRY HOIST LOADING A BIOMASS BASKET INTO THE RETORT

1 Ton	10+ Yrs	2-Stage	In-House
OUTPUT CAPACITY	CONTINUOUS OPERATION*	CONDENSATE RECOVERY	MANUFACTURED

i.

Premium Construction

High-grade alloy steel with refractory lining, engineered for 10+ years of continuous operation when run as per guidelines.

ii.

Condensate Collection

Dual-stage condensate collection recovers wood vinegar and smoke oil, creating additional revenue from every production cycle.

iii.

Advanced Design

Advanced design locks carbon content in and ensures biomass is effectively converted into high-grade charcoal.

* When operated and maintained as per ChaarGold guidelines.

KEY FEATURES

Built Around Performance

	FEATURE	WHAT IT DELIVERS
1	Precision-engineered construction	Tight fabrication and in-house quality control, batch after batch.
2	Optimized thermal efficiency	Precise temperature control and efficient heat use cut fuel cost.
3	Sustainable biomass conversion	Turns biomass residues into carbon products, with low emissions.
4	High charcoal & biochar yields	Carbon is locked in for higher yields and a high-grade product.
5	Recovery of valuable by-products	Dual-stage condensation recovers wood vinegar and smoke oil.
6	Robust & durable design	Alloy steel with refractory lining; 10+ years of continuous operation.
7	Scalable production capacities	Start with a 1 ton unit; grow into a multi-kiln line as volumes rise.
8	Continuous process optimization	Continuous refinement and support to improve yield and quality.

THE PROCESS

From Biomass to Value

1

Charge

Biomass is loaded in a basket by gantry hoist, keeping handling quick and clean.

2

Carbonize

Heated with limited oxygen, biomass converts to carbon-rich solids and vapours.

3

Condense

Vapours pass through two condensation stages to recover wood vinegar and smoke oil.

4

Discharge

Finished charcoal or biochar is removed, ready for grading, briquetting or use.

OUTPUT

Products Generated Through Our Kilns

PRODUCT I

Lump Charcoal

Premium-quality charcoal with high fixed carbon content, long burn duration and low ash generation.

Briquettes · Pillow · Hexagonal · Honeycomb

PRODUCT II

Biochar

Carbon-rich material engineered for agriculture, carbon sequestration, soil enhancement and environmental applications.

Agriculture · Carbon removal · Soil health

PRODUCT III

Wood Vinegar

A valuable pyroligneous acid used in agriculture, composting, odor management and soil health improvement.

Smoke oil recovered as a further by-product

TURNKEY SOLUTIONS

Beyond the Kiln

ChaarGold offers complete turnkey pyrolysis plant solutions — one accountable partner from the first conversation to a fully running plant.

01	Project consultation and feasibility assessment
02	Kiln design and manufacturing
03	Installation and commissioning
04	Operator training
05	Production optimization
06	Technical support and after-sales service
07	Process integration for biochar and charcoal production

WHY CHAARGOLD

More Than Equipment

Our kilns are the result of years of practical manufacturing experience and engineering innovation. Every system is designed to help clients achieve higher production efficiency, superior product quality and long-term operational reliability.

With installations across multiple industries and applications, ChaarGold has earned a reputation for dependable pyrolysis technology, tailored to real-world operating conditions.

From a single 1 ton kiln to a multi-kiln production line, we build systems that grow with your business.

INDUSTRIES WE SERVE

I Agriculture & Biochar Producers	II Charcoal Manufacturers	III Industrial Carbon Producers	IV Waste-to-Value Projects
V Carbon Credit Developers	VI Forestry & Biomass Processing	VII Research & Development Institutions	VIII Activated Charcoal Industries



LEADERSHIP

Prem Raj

FOUNDER & CHIEF TECHNOLOGY OFFICER

Prem Raj is the Founder and Chief Technology Officer of CHAARGOLD, bringing over 15 years of experience in sustainable energy, biomass conversion and carbon solutions. An engineer, innovator, patent holder and passionate agriculturist, he combines technical expertise with practical industry knowledge to develop advanced pyrolysis technologies and high-performance carbon products.

With global exposure across diverse markets, he is also a successful hotelier and brand-builder, driven by a vision to create sustainable businesses that deliver lasting value for customers, communities and the environment.

> *Innovation is not just about creating new technologies; it is about transforming challenges into sustainable opportunities that create lasting value for people and the planet.*
>
> — Prem Raj